NBA/NHL Expected Points and Points Over Expected Analysis

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Title and Abstract

This proposal analyzes shooting efficiency in the NBA and NHL using probabilistic models and residual-based metrics. We estimate shot-level scoring probability with NBA xFG (logistic regression) and NHL xG (LASSO plus GAM diagnostics), then quantify deviations via Points/Goals Over Expected (POE) and the Shot Difficulty Index (SDI). The draft includes two NBA figures from the existing pipeline and two NHL figures derived from teammate data, each with concise interpretation. Deliverables are a reproducible R Markdown workflow, a balanced cross-league analysis, and a submission-ready 2-3 page PDF.

Introduction

Expected-value modeling separates shot quality from outcomes. We analyze NBA optical-tracking data from 2014-15 through 2025-26 and NHL MoneyPuck data from 2007-2024 to enable fair cross-league comparisons across roles, volume, and shot environments.

Table 1: Sample NBA Player Summary (2025-26)

Player	Shots	FG%	Total POE	POE/100
AJ Green	391	0.4398977	73.166894	18.7127606
Aaron Gordon	273	0.5091575	14.497316	5.3103721
Aaron Nesmith	370	0.3891892	-30.114935	-8.1391715
Aaron Wiggins	363	0.4435262	-3.078856	-0.8481698
Ace Bailey	518	0.4498069	-42.170340	-8.1409924
Ajay Mitchell	456	0.4868421	22.561261	4.9476449

Methods

We model NBA xFG with logistic regression and NHL xG with LASSO using shot-context features (distance, angle, context, and game-state proxies). NHL shot-level xGoal probabilities are used directly in residual comparisons with observed goals. We then compute residual-based metrics to assess performance relative to expectation at the player level and apply GAM diagnostics to validate non-linear spatial effects. To aid interpretation, both pipelines frame efficiency through SDI-like metrics, evaluating difficulty-adjusted performance rather than raw conversion rates.

For the NHL visuals, data are loaded via NHL_CSV_PATH with fallback paths. The code fails fast if data are unavailable, preserving portability and avoiding hard-coded machine paths.

Expected Outcomes

NBA Visual Evidence

Figure 1 shows the expected inverse relationship between SDI and xFG%, confirming that shot context drives modeled efficiency in the NBA sample. Figure 2 highlights value dispersion at similar salary levels, motivating POE per salary context rather than salary-only evaluation. Positive outliers indicate players returning more scoring value than their contract tier suggests.

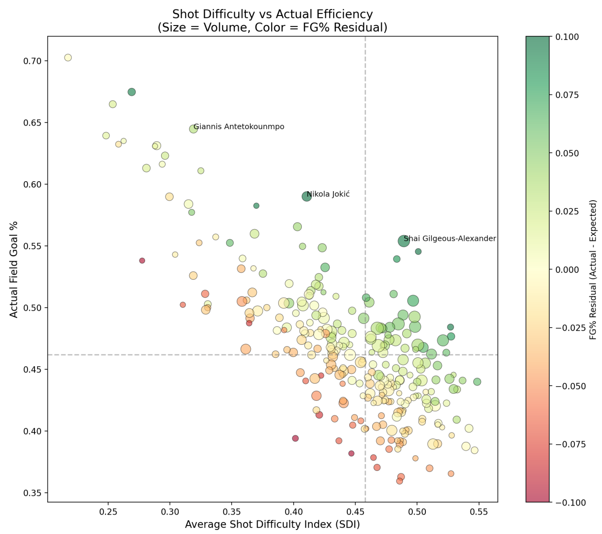

Figure 1: Figure 1: NBA Shot Difficulty Index vs Expected FG% (2025-26)

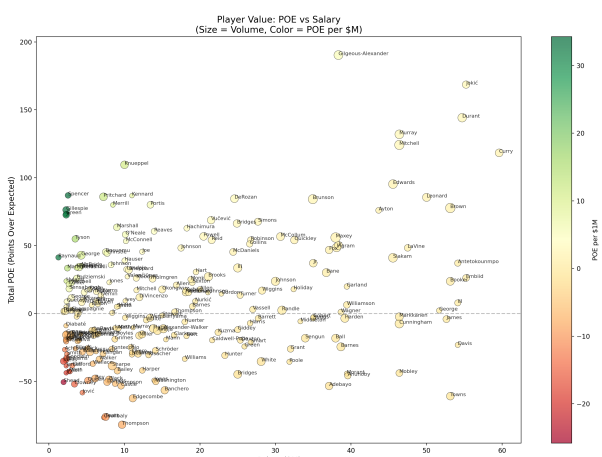

Figure 2: Figure 2: NBA Points Over Expected vs Salary (2025-26)

NHL Visual Evidence

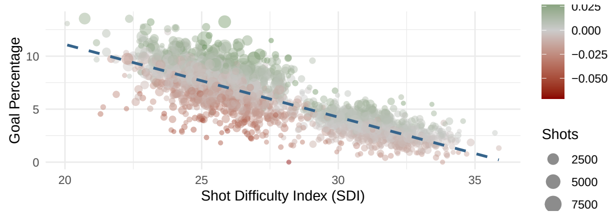

Figure 3: Figure 3: NHL Shot Difficulty Index vs Goal Percentage (players with >=100 shots)

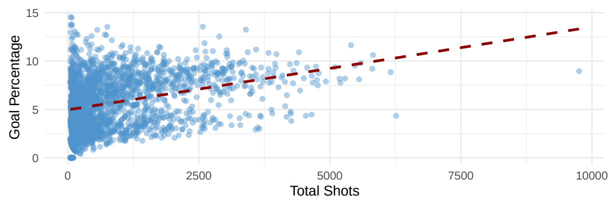

Figure 4: Figure 4: NHL Shot Volume vs Goal Percentage (players with >=50 shots)

Figure 3 mirrors the NBA pattern: players taking harder shots trend toward lower conversion rates. Figure 4 shows that higher NHL shot volume does not guarantee higher finishing efficiency. The weak linear trend supports evaluating players on shot quality and residual performance, not volume alone.

Figure 1 (NBA SDI vs xFG) and Figure 3 (NHL SDI vs goal percentage) show that higher shot difficulty lowers scoring probability. Figures 2 and 4 illustrate why outcomes require context: volume and salary alone do not explain efficiency outliers. Together, they justify a shared expected-points framework across leagues.

Group Considerations

Our three-person team split work across data access, modeling, and communication deliverables so each league could be developed in parallel and then integrated in one document.

Task	Kanaan	Kaleb	Drianna
Data Collection (NBA/NHL)	0%	50%	50%
Feature Engineering + SDI	0%	50%	50%
Modeling (xFG/xG, diagnostics)	30%	40%	30%
Figures + Proposal Writing	33%	33%	33%